QDB SME Relief Portal — Product Requirements Document

Product: QDB SME Relief Portal **Version**: 1.0 **Date**: March 3, 2026 **Classification**: Confidential — QDB Internal Use Only
Author: ConnectSW Product Manager **Status**: Final Draft — Ready for Architecture Review

Table of Contents

1. Product Overview
2. User Personas
3. User Stories and Acceptance Criteria
4. Functional Requirements
5. Non-Functional Requirements
6. Mermaid Diagrams
7. Integration Specifications
8. Acceptance Criteria Summary
9. Site Map
10. Out of Scope
11. Open Questions and Assumptions Requiring Resolution
12. Traceability Matrix

1. Product Overview

1.1 Product Name

QDB SME Relief Portal

1.2 One-Line Description

An emergency financing portal that enables Qatari SMEs affected by geopolitical disruption to apply for relief under the National Relief and Guarantee Program (NRGP), with automated eligibility checking, WPS-validated salary verification, and direct routing into QDB's Dynamics CRM case management system.

1.3 Business Context

The problem: Qatar's SME sector is experiencing acute cash flow stress due to the Iran-Israel geopolitical conflict. Trade routes are disrupted, supply chains fragmented, and regional demand suppressed. Affected SMEs face simultaneous fixed obligations: salary payments legally mandated under Qatar's Wage Protection System (WPS) and commercial lease obligations. Without emergency financing, these businesses face WPS violations (QAR 6,000 per employee per late cycle), lease defaults, and closure.

The program: The National Relief and Guarantee Program (NRGP) is an existing QDB financing instrument with board-level approval, an active beneficiary list from prior activations (COVID-19), and pre-negotiated guarantee structures. QDB is not creating a new product — it is digitising delivery of an approved, standing program under an emergency activation mandate.

The portal's purpose: A structured digital workflow replaces a manual, paper-heavy process that takes 15–30 business days to process a single application. The portal authenticates applicants via Tawtheeq (Qatar's National Authentication Service), retrieves verified company data from the Ministry of Commerce and Industry (MOCI), evaluates eligibility automatically against NRGP rules, validates salary obligations against WPS payroll records, and routes disbursement requests — automatically for returning NRGP beneficiaries, or for manual QDB review for new applicants — into Microsoft Dynamics CRM.

Urgency: Every month of delay exposes SMEs to WPS violations and irreversible business harm. A 6–8 week delivery target is required.

1.4 Target Users

- SME owners or authorized signatories of Qatar-registered companies with active Tawtheeq accounts
- QDB Relationship Managers reviewing manual disbursement cases in Dynamics CRM
- QDB Administrators managing eligibility rules, the NRGP list, and program lifecycle

1.5 Success Metrics

Metric	Target
Auto-disbursement path end-to-end time	< 1 business day from submission to CRM case creation
Manual review path processing time	< 5 business days from submission to QDB decision
Document re-submission rate	< 10% (down from 40–60% manual)
Eligible applications processed per week	500+ (up from ~20 manual)
NAS authentication success rate	> 98%
MOCI API success rate	> 99.5%
WPS validation completion rate	> 95% of submitted applications
CRM case creation failure rate	< 0.1%
Portal uptime during program window	> 99.5%
Duplicate application detection rate	100%
Audit trail completeness	100% of application steps logged
Applicant satisfaction score	> 4.0 / 5.0

2. User Personas

Persona 1: Khalid — SME Owner (Primary Applicant)

- **Role**: Owner and authorized signatory of a mid-size Qatari trading company (25–60 employees). Also represents the financial controller archetype at larger SMEs who delegates submission.
- **Demographics**: Qatari national or long-term resident. Operates primarily in Arabic. May have limited experience with government digital portals.
- **Goals**:
  - Access emergency financing before the next salary cycle to avoid WPS violations
  - Understand eligibility quickly without waiting days for a QDB response
  - Submit all required documents in a single sitting without needing to revisit
  - Track application status without calling QDB
- **Pain Points (current)**:
  - No clarity on whether his business qualifies before investing hours in paperwork
  - Multiple back-and-forth email rounds with QDB for missing documents
  - No visibility on application status; must phone QDB to get updates
  - English-only portal creates friction for Arabic-preferring users
- **Usage Context**: Applies once, via desktop or laptop, during business hours. Returns to the portal 1–3 times to check status and receive notifications. Motivated by urgency — a salary deadline is approaching.
- **Key Need**: A fast, clear, Arabic-language portal that tells him upfront whether he qualifies, guides him to upload the right documents the first time, and then keeps him informed.

Persona 2: Fatima — QDB Relationship Manager

- **Role**: Senior RM responsible for reviewing manual disbursement cases for SMEs not on the existing NRGP list. Reviews cases in Dynamics CRM. Makes recommendations to Credit Risk for borderline cases.
- **Goals**:
  - Access complete application information (company data, eligibility result, documents, WPS validation) in one CRM view without going back to the applicant
  - Process cases quickly to meet program throughput targets
  - Identify and escalate cases with discrepancies (WPS mismatches, incomplete documents)
- **Pain Points (current)**:
  - Currently receives email bundles of scanned documents; manual data entry into CRM
  - Frequently contacts applicants to request missing items
  - No structured audit trail to rely on for compliance reporting
- **Usage Context**: Uses Dynamics CRM daily. Does not log into the portal directly — views portal-created case records and linked documents.
- **Key Need**: CRM cases that arrive with all information pre-populated, correctly routed, and flagged for any WPS or document discrepancies requiring her attention.

Persona 3: Mohammed — QDB Admin / Program Manager

- **Role**: QDB Operations lead responsible for the NRGP activation. Manages eligibility rule configuration, uploads updated NRGP beneficiary lists, monitors application volumes, and enforces the program lifecycle (open/close).
- **Goals**:
  - Update eligibility rules if QDB policy changes during the program window — without requesting a code deployment
  - See real-time dashboard of application volumes, statuses, and processing bottlenecks
  - Upload a new NRGP beneficiary list when QDB operations provides an updated version
  - Close the program on the agreed end date and confirm no new applications enter after that point
- **Pain Points (current)**:
  - Eligibility changes require IT involvement and delay; no admin interface
  - No consolidated view of the program — must query multiple spreadsheets and email trails
  - No automated program sunset; risk of unauthorized applications after the window closes
- **Usage Context**: Logs into a QDB admin section of the portal. Uses it primarily for configuration, monitoring, and lifecycle management — not applicant-facing tasks.
- **Key Need**: An admin interface that gives him full control of program parameters without requiring engineering.

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3. User Stories and Acceptance Criteria

Authentication Flow (BN-001)

US-01 — Tawtheeq Login

> As an SME owner, I want to log in with my existing Tawtheeq account so that I do not need to create a new credential to access the QDB relief portal.

Priority: P0 **Maps to**: BN-001, FR-001

Acceptance Criteria:

- **Given** a user visits the portal home page and clicks "Apply for Relief", **When** they are redirected to NAS/Tawtheeq and complete authentication successfully, **Then** the portal receives the QID claim, establishes a session, and lands the user on the application start page.
- **Given** a user fails NAS authentication (wrong credentials, locked account), **When** they return to the portal after a failed NAS attempt, **Then** the portal displays a clear message: "Authentication failed. Please retry Tawtheeq login or contact QDB Operations at [phone]." No partial session is created.
- **Given** a user does not have a Tawtheeq account, **When** they land on the portal and select "I don't have a Tawtheeq account", **Then** the portal displays guided instructions for NAS registration and a QDB Operations contact for assisted onboarding.
- **Given** NAS/Tawtheeq is temporarily unavailable, **When** the portal attempts to initiate the OIDC flow and receives a timeout or error, **Then** the portal displays: "The national authentication service is temporarily unavailable. Please try again in 30 minutes or contact QDB Operations." No session is created and no sensitive data is exposed.

US-02 — Session Management and Logout

> As an SME owner, I want my portal session to remain active while I am completing my application and to expire automatically after inactivity so that my company data is not left exposed on a shared device.

Priority: P0 **Maps to**: BN-001, FR-001

Acceptance Criteria:

- **Given** an authenticated user is active in the portal, **When** they perform any action (page navigation, form input, file upload), **Then** the session timer resets and they remain authenticated.
- **Given** an authenticated user has been idle for 30 minutes, **When** the inactivity timer expires, **Then** the portal displays a "Session expiring in 2 minutes" warning with an option to extend. If not extended, the session is terminated, the user is redirected to the login page, and application progress saved to that point is persisted.
- **Given** an authenticated user clicks "Logout", **When** the logout action is confirmed, **Then** the portal invalidates the session token, clears all session data from the browser, logs the logout event to the audit trail, and redirects to the portal home page.
- **Given** a user attempts to access a protected page without a valid session, **When** the request is made, **Then** the user is redirected to the NAS login flow and returned to their intended page after successful authentication.

Company Verification Flow (BN-002)

US-03 — Enter CR Number and Fetch MOCI Data

> As an SME owner, I want to enter my Commercial Registration number and have the system retrieve my verified company details from MOCI so that I do not need to manually type company information that could introduce errors.

Priority: P0 **Maps to**: BN-002, FR-002

Acceptance Criteria:

- **Given** an authenticated user enters a valid, active 10-digit CR number, **When** they click "Verify Company", **Then** the portal queries the MOCI API within 3 seconds and displays: company name (Arabic and English), CR status (active), sector classification, and registration date.
- **Given** an authenticated user enters a CR number that does not exist in MOCI, **When** they click "Verify Company", **Then** the portal displays: "This CR number was not found in MOCI records. Please verify the number and try again. If the issue persists, contact MOCI at [link]." The field is cleared and ready for re-entry.
- **Given** an authenticated user enters a CR number for a company with inactive, suspended, or cancelled status, **When** MOCI returns the record, **Then** the portal displays: "Your company's CR status is [status] and is not eligible for NRGP relief. If this is incorrect, please contact MOCI." The application cannot proceed.
- **Given** the MOCI API is unavailable at the time of lookup, **When** the API request times out or returns an error, **Then** the portal displays: "Company verification is temporarily unavailable. Please try again in 10 minutes. Your progress

has been saved." The portal does not allow manual entry of company details as a bypass.

- **Given** a valid CR number is fetched successfully, **When** the user reviews the company details, **Then** all displayed fields are read-only; the user cannot modify MOCI-sourced company data.

US-04 — Confirm Authorized Signatory Status

> As an SME owner, I want the system to confirm that I am an authorized signatory for the CR I have entered so that the application is legally valid and I am not inadvertently submitting on behalf of a company I do not represent.

Priority: P0 **Maps to**: BN-002, BN-008, FR-002

Acceptance Criteria:

- **Given** a user has successfully fetched company data for CR-X, **When** the portal cross-references the authenticated QID against the MOCI signatory list for CR-X, **Then** if the QID matches an authorized signatory, the user proceeds to the eligibility step without additional friction.
- **Given** the MOCI signatory data does not include the authenticated QID for CR-X, **When** the check fails, **Then** the portal displays: "You are not listed as an authorized signatory for this company in MOCI records. If you are authorized, please contact QDB Operations with your authorization documentation." The application is blocked.
- **Given** MOCI does not return signatory data (data gap), **When** the check cannot be completed automatically, **Then** the portal prompts the user to make a statutory declaration: "I confirm that I am legally authorized to submit this application on behalf of [Company Name]." The declaration is captured in the audit trail with timestamp and QID. The portal proceeds but flags the case for QDB authorization verification.
- **Given** any company authorization event occurs, **When** the result is determined, **Then** the audit trail records: QID, CR number, check result, data source, timestamp.

Eligibility Check Flow (BN-003)

US-05 — Automatic Eligibility Evaluation

> As an SME owner, I want the system to automatically evaluate my company's eligibility for NRGP relief immediately after company verification so that I know whether to proceed before investing time in document preparation.

Priority: P0 **Maps to**: BN-003, FR-003

Acceptance Criteria:

- **Given** company verification is complete and the authorized signatory check passes, **When** the eligibility engine runs, **Then** all six eligibility criteria (EC-001 through EC-006) are evaluated in sequence and a pass/fail result is produced within 5 seconds.
- **Given** the eligibility engine produces an ELIGIBLE result, **When** the result is displayed, **Then** the user sees a clear "Eligible" confirmation and is automatically advanced to the NRGP list check step.

- **Given** the eligibility engine produces an INELIGIBLE result, **When** the result is displayed, **Then** the portal shows each failed criterion with a plain-language reason in both Arabic and English, a reference to the relevant NRGP policy section, and contact details for QDB to discuss borderline cases.
- **Given** any eligibility evaluation, **When** the result is determined, **Then** the audit trail records: CR number, evaluation timestamp, criteria snapshot (the exact rule values in effect at evaluation time), and each criterion result (pass/fail with reason).

US-06 — View Eligibility Result with Reason Codes

> As an SME owner, I want to see a detailed eligibility result with specific reason codes so that I understand exactly why my application was accepted or rejected and what action I can take.

Priority: P0 **Maps to**: BN-003, BN-009, FR-003

Acceptance Criteria:

- **Given** an ineligible result is returned, **When** the result page renders, **Then** each failed criterion is presented with:
  - A short reason code (e.g., ELIG-003: Company age below 12 months)
  - A plain-language explanation in both Arabic and English
  - The specific MOCI or WPS data value that caused the failure
  - A recommended action if one exists
- **Given** the eligibility result is displayed, **When** the user refreshes or returns to the page, **Then** the result persists for the duration of the session and is accessible via the application status page.
- **Given** an eligible result is returned, **When** the result page renders, **Then** the page shows a summary of criteria passed and moves the user forward automatically — it does not require a manual "Continue" click on the eligibility result page.

US-07 — Admin Configures Eligibility Criteria

> As a QDB Admin, I want to update eligibility criteria through the admin interface without requiring a code deployment so that policy changes can be reflected on the same day QDB Credit Risk approves them.

Priority: P2 **Maps to**: BN-016, FR-003

Acceptance Criteria:

- **Given** a QDB Admin is authenticated to the admin dashboard, **When** they navigate to "Eligibility Rules", **Then** they see a list of all active criteria (EC-001 through EC-007) with their current values and effective dates.
- **Given** a QDB Admin modifies a criterion value (e.g., changes minimum company age from 12 to 6 months), **When** they save the change, **Then** the new rule takes effect for all subsequent eligibility evaluations. Evaluations already completed are not retroactively affected. The change is recorded in the audit trail with the admin's identity and timestamp.
- **Given** a QDB Admin attempts to delete a mandatory criterion (EC-001, EC-002, EC-004), **When** they attempt the deletion, **Then** the system prevents the action and displays: "This criterion is mandatory under NRGP policy and

cannot be removed."

- **Given** any criterion change is saved, **When** the save is confirmed, **Then** a confirmation email is sent to the admin's QDB email address listing the change details for record-keeping.

NRGP Check and Disbursement Routing Flow (BN-004, BN-005)

US-08 — NRGP List Match — Auto Disbursement Path

> As a QDB Operations Officer, I want auto-disbursement CRM cases to be created automatically for eligible NRGP-listed companies so that pre-vetted applicants receive relief with minimal QDB intervention.

Priority: P0 **Maps to**: BN-004, FR-004, FR-005

Acceptance Criteria:

- **Given** an eligible company's CR number is found in the NRGP beneficiary list, **When** the NRGP check completes, **Then** the portal creates a CRM case of type `auto_nrgp` with status `pending_disbursement` and the CRM Case ID is returned to the portal within 5 seconds.
- **Given** the auto-disbursement CRM case is created, **When** the case is queried in Dynamics CRM, **Then** it contains: company name, CR number, QID, NRGP list match confirmation, eligibility result snapshot, timestamp, and a placeholder for document IDs (to be populated after upload).
- **Given** the CRM case ID is returned, **When** the portal displays the next step to the applicant, **Then** the applicant sees their CRM Case ID as their application reference number and is directed to the document upload step.
- **Given** the CRM API is unavailable at the time of case creation, **When** the API returns an error or times out, **Then** the portal retries up to 3 times with exponential back-off. If all retries fail, the application state is stored in the portal database and a QDB Operations alert is triggered. The applicant is shown: "Your case is being registered. You will receive an email with your case reference within 2 hours."

US-09 — NRGP List No Match — Manual Review Path

> As a QDB Relationship Manager, I want manual review CRM cases to arrive with all application data and supporting context pre-populated so that I can make an informed decision without contacting the applicant for basic information.

Priority: P0 **Maps to**: BN-005, FR-004, FR-005

Acceptance Criteria:

- **Given** an eligible company's CR number is NOT found in the NRGP beneficiary list, **When** the NRGP check completes, **Then** the portal creates a CRM case of type `manual_review` with status `pending_review` and the CRM Case ID is returned to the portal.
- **Given** the manual review CRM case is created, **When** a QDB Relationship Manager opens it in Dynamics CRM, **Then** the case shows: company name, CR number, MOCI company data (sector, registration date, shareholder summary),

eligibility criteria result for each criterion, WPS validation result, and links to all uploaded documents.

- **Given** the CRM case ID is returned to the portal, **When** the applicant views their application summary, **Then** they see a message explaining: "Your application is assigned to a QDB Relationship Manager for review. Expected review time is 5 business days. Your case reference is [CASE-ID]."

Document Upload Flow (BN-006, BN-007)

US-10 — Upload Salary Payment Evidence

> As an SME owner, I want to upload salary payment evidence so that QDB has proof of my payroll obligations and can calculate the correct salary relief amount.

Priority: P0 **Maps to**: BN-006, FR-006

Acceptance Criteria:

- **Given** an applicant is on the document upload step, **When** they upload a salary payment evidence file (bank transfer confirmation, payroll run report), **Then** the portal accepts PDF, JPEG, PNG, and TIFF formats up to 10 MB per file. Files larger than 10 MB are rejected with a clear size error.
- **Given** a salary evidence file is uploaded successfully, **When** the upload completes, **Then** the portal displays a thumbnail or filename, an upload status indicator ("Uploaded"), and a checksum confirmation. The file is stored in encrypted document storage under `{case_id}/salary_evidence/`.
- **Given** a salary evidence file upload fails (network error, unsupported format), **When** the failure occurs, **Then** the portal displays a specific error message (e.g., "File format not supported. Please upload PDF, JPEG, PNG, or TIFF") and allows immediate retry without losing other uploaded files.
- **Given** an applicant uploads multiple salary evidence files, **When** the upload step is complete, **Then** all files appear in a list with individual status indicators. The applicant can remove any file before final submission.

US-11 — Upload WPS File for Salary Validation

> As an SME owner, I want to upload my WPS payroll file so that the system can cross-validate my salary claims against Ministry of Labor records.

Priority: P0 **Maps to**: BN-007, FR-006, FR-007

Acceptance Criteria:

- **Given** an applicant is on the document upload step, **When** they upload a WPS file, **Then** the portal accepts the WPS-standard CSV format and rejects any other format with a message: "Please upload the WPS file in the standard Ministry of Labor CSV format."
- **Given** a valid WPS file is uploaded, **When** the file is processed, **Then** the portal parses the file, displays a summary: employee count, total monthly payroll figure (last 90 days), and earliest/latest payment date in the file. The applicant can confirm or flag a discrepancy before proceeding.

- **Given** the WPS file data differs from the live WPS API query by more than 10%, **When** the discrepancy is detected, **Then** the CRM case is flagged with a `wps_discrepancy` note and the applicant is shown a warning (not a block): "There is a discrepancy between your uploaded WPS file and Ministry of Labor records. Your application will continue but will be flagged for QDB review."
- **Given** no WPS file is provided by the applicant, **When** the system attempts WPS validation via the WPS API, **Then** if the API is available, validation proceeds automatically using API data. If unavailable, the case is flagged for manual salary verification and the applicant is notified.

US-12 — Upload Rent Payment Evidence

> As an SME owner, I want to upload rent payment evidence so that QDB can calculate the rent relief component of my application.

Priority: P0 **Maps to**: BN-006, FR-006

Acceptance Criteria:

- **Given** an applicant is on the document upload step, **When** they upload rent evidence (lease agreement, rent payment receipts, landlord invoice), **Then** the portal accepts PDF, JPEG, PNG, and TIFF formats up to 10 MB per file, stores files under `{case_id}/rent_evidence/`, and displays upload confirmation.
- **Given** an applicant uploads both a lease agreement and payment receipts, **When** both are uploaded, **Then** both appear in the rent evidence section with individual status indicators.
- **Given** an applicant indicates they do not have a commercial lease (e.g., home-based business), **When** they skip the rent evidence section, **Then** the portal allows the application to proceed and sets the rent relief component to QAR 0 on the CRM case.

US-13 — Upload Company Documents

> As an SME owner, I want to upload my commercial registration copy and business licenses so that QDB has the official documentation to support my application.

Priority: P0 **Maps to**: BN-006, FR-006

Acceptance Criteria:

- **Given** an applicant is on the document upload step, **When** they upload a CR copy (scanned PDF or JPEG), **Then** the portal validates the format, stores the file under `{case_id}/company_docs/`, and displays a confirmation. Files under 10 MB are accepted.
- **Given** an applicant uploads additional business licenses, **When** they upload up to 5 license files, **Then** all files are accepted, individually listed, and linked to the CRM case.
- **Given** the total document upload package for a single application exceeds 50 MB, **When** the applicant attempts to add another file that would breach the limit, **Then** the portal displays: "Your total upload package has reached the 50 MB limit. Please compress files or contact QDB Operations for assistance."

US-14 — View Upload Status and Validation Results

> As an SME owner, I want to see the status of each uploaded document so that I know which uploads were accepted and which need to be retried before I submit.

Priority: P0 **Maps to**: BN-006, BN-011, FR-006

Acceptance Criteria:

- **Given** a document has been uploaded successfully, **When** the applicant views the upload section, **Then** each successfully uploaded document shows a green checkmark, the filename, upload timestamp, and file size.
- **Given** a document has failed validation (wrong format, too large, virus detected), **When** the applicant views the upload section, **Then** the failed document shows a red error icon, the filename, and a specific error message. The "Submit Application" button is disabled until all required documents have passed validation.
- **Given** an optional document category has no uploads, **When** the applicant views the upload section, **Then** the optional category shows as "Not provided" with a neutral indicator. The application can still be submitted.
- **Given** all required documents are uploaded and validated, **When** the applicant views the upload section, **Then** the "Submit Application" button becomes active and a checklist summary is displayed.

WPS Validation Flow (BN-007)

US-15 — System Validates WPS File Format and Data

> As a QDB Credit Officer, I want the portal to validate salary obligations against WPS data automatically so that relief amounts are anchored to Ministry of Labor payroll records and cannot be inflated.

Priority: P0 **Maps to**: BN-007, FR-007

Acceptance Criteria:

- **Given** the portal initiates a WPS API query for a given CR number, **When** the WPS system responds, **Then** the portal retrieves: employee count, total monthly payroll (last 90 days), per-month breakdown, and payment dates. This data is no older than 90 days (BR-010).
- **Given** WPS API data is returned, **When** the data is compared to the applicant's declared salary relief amount, **Then** if the declared amount is within 10% of the WPS figure, validation passes. If the discrepancy exceeds 10%, a `wps_discrepancy` flag is set on the CRM case.
- **Given** the WPS API returns no records for the CR number, **When** the query completes, **Then** the portal flags the case with `wps_no_records` and sets a manual salary verification requirement on the CRM case. The application is not blocked.
- **Given** WPS validation completes (pass, flag, or no records), **When** the result is written, **Then** the audit trail records: CR number, WPS query timestamp, WPS response summary, declared amount, discrepancy percentage (if applicable), and validation outcome.

US-16 — View WPS Validation Results and Discrepancies

> As an SME owner, I want to see the WPS validation result so that I understand how my declared salary amount compares to Ministry of Labor records and can address any discrepancy before final submission.

Priority: P1 **Maps to**: BN-007, BN-009, FR-007

Acceptance Criteria:

- **Given** WPS validation has run, **When** the applicant reaches the WPS result step, **Then** they see: WPS-validated employee count, WPS-validated total payroll (last 3 months), their declared salary amount, and a comparison showing whether the amounts match or differ.
- **Given** a WPS discrepancy flag is set, **When** the applicant views the WPS result, **Then** the portal shows a non-blocking warning with a plain-language explanation: "There is a difference between your declared salary amount and Ministry of Labor records. Your application will proceed but will be reviewed by a QDB officer."
- **Given** WPS validation passes without discrepancy, **When** the applicant views the WPS result, **Then** the portal shows a confirmation: "Your salary obligations have been verified against Ministry of Labor records."

Application Status Flow (BN-009)

US-17 — View Application Status and History

> As an SME owner, I want to see my application status and a log of key events so that I know exactly where my application is in the process without needing to contact QDB.

Priority: P1 **Maps to**: BN-009, FR-008

Acceptance Criteria:

- **Given** an applicant has submitted an application, **When** they log into the portal and navigate to "My Application", **Then** they see: case reference number, current status, status change history (chronological list of status events with timestamps), and the expected next step.
- **Given** the application status is updated in Dynamics CRM (by a QDB RM), **When** the applicant next views their status page, **Then** the portal reflects the updated CRM status within 5 minutes of the CRM update. Status labels are displayed in both Arabic and English.
- **Given** a status of "Rejected" is displayed, **When** the applicant views their status, **Then** the portal shows the rejection reason (provided by the QDB RM in CRM) and next steps (e.g., appeal contact).
- **Given** a status of "Disbursed" is displayed, **When** the applicant views their status, **Then** the portal shows the disbursement confirmation and directs the applicant to contact their QDB relationship manager for bank transfer details.

US-18 — Receive Notifications on Status Changes

> As an SME owner, I want to receive email and SMS notifications when my application status changes so that I am informed without having to poll the portal.

Priority: P1 **Maps to**: BN-012, FR-009

Acceptance Criteria:

- **Given** an application status changes in the system (submitted, under review, approved, rejected, disbursed), **When** the status change event fires, **Then** the portal sends a notification via email and (if mobile number provided) SMS within 5 minutes of the status change.
- **Given** a notification is sent, **When** the notification is delivered, **Then** the message includes: applicant name, company name, case reference, new status, plain-language description of what the status means, and a link to the status page in the portal.
- **Given** a notification is sent in Arabic, **When** the applicant's language preference is Arabic, **Then** the email and SMS are delivered in Arabic with correct RTL text direction in the email body.
- **Given** a notification delivery fails (invalid email, network error), **When** the failure is detected, **Then** the failure is logged, a retry is scheduled after 10 minutes, and after 3 failed attempts the case is flagged for QDB Operations manual outreach.

QDB Management (BN-013)

US-19 — QDB Dashboard — View All Applications

> As a QDB Operations Manager, I want a dashboard showing all applications with their current status, submission time, and processing timelines so that I can manage team capacity and report program progress to the QDB board.

Priority: P1 **Maps to**: BN-013, FR-011

Acceptance Criteria:

- **Given** a QDB Admin is authenticated to the admin dashboard, **When** they view the Applications Overview panel, **Then** they see a table of all applications with columns: Case ID, Company Name, CR Number, Submission Date, Current Status, CRM Case Type (auto/manual), and Days Since Submission.
- **Given** the dashboard is loaded, **When** an admin filters by status (all, under review, approved, rejected, disbursed), **Then** the table filters in real time and displays a count of records matching the filter.
- **Given** the dashboard displays application data, **When** an admin views the KPI summary panel, **Then** they see: total applications submitted, count by status, average processing time (auto vs. manual paths), document re-submission rate, and WPS discrepancy rate.
- **Given** an admin clicks on a specific application row, **When** the detail view opens, **Then** they see the full application timeline, eligibility result, NRGP list match result, document upload status, WPS validation result, and a link to the corresponding CRM case.

US-20 — QDB Admin — Upload and Update NRGP List

> As a QDB Admin, I want to upload an updated NRGP beneficiary list through the admin interface so that the system immediately uses the latest list for all subsequent applications.

Priority: P1 **Maps to**: BN-013, FR-004, FR-011

Acceptance Criteria:

- **Given** a QDB Admin is on the "NRGP List Management" admin page, **When** they upload a new NRGP list file (CSV format, CR numbers, one per row), **Then** the portal validates the file format, counts the records, displays a preview of the first 10 rows, and asks for confirmation before activating.
- **Given** the admin confirms the new NRGP list upload, **When** the list is activated, **Then** all subsequent NRGP lookup steps use the new list. Applications already submitted are not retroactively affected. The activation event is recorded in the audit trail with the admin's identity, timestamp, record count (old and new), and a checksum of the uploaded file.
- **Given** the uploaded file has a format error (missing header, invalid CR format), **When** the validation runs, **Then** the portal rejects the file, displays the specific error and row number, and the existing NRGP list remains active.
- **Given** a QDB Admin views the NRGP List Management page, **When** the page loads, **Then** they can see the active list's upload date, record count, and the identity of the admin who last uploaded it.

US-21 — QDB Admin — Configure Eligibility Rules

> As a QDB Admin, I want to modify eligibility rule parameters without a code deployment so that policy changes are reflected immediately.

Priority: P2 **Maps to**: BN-016, FR-003, FR-011

Acceptance Criteria: See US-07 — Acceptance criteria are shared. This story is the admin-facing complement.

Audit and Compliance (BN-008, BN-018)

US-22 — Tamper-Evident Audit Trail for All Actions

> As a QDB Compliance Officer, I want every application action, system decision, and data source to be logged in a tamper-evident audit record so that the NRGP program can be fully audited by government authorities.

Priority: P0 **Maps to**: BN-008, FR-008

Acceptance Criteria:

- **Given** any application step occurs (login, CR lookup, eligibility check, NRGP lookup, CRM case creation, document upload, WPS validation, status change, notification send), **When** the step completes (success or failure), **Then** an

audit record is written containing: event type, timestamp (UTC), actor QID or system identifier, input data summary (not raw PII), output/result, data source identifier (MOCI, WPS, NRGP list version, CRM case ID), and application session ID.

- **Given** an audit record is written, **When** it is stored, **Then** the record is written to an append-only table. Updates and deletes on audit records are prevented at the database constraint level. Any attempt to modify an audit record triggers an alert.
- **Given** a QDB Compliance Officer requests an audit export for a specific application, **When** they use the admin export function, **Then** they receive a structured export (JSON or CSV) of all audit events for that application in chronological order, including all system decisions with the criteria snapshot in effect at decision time.
- **Given** the audit log write fails for any reason, **When** the failure is detected, **Then** the application step that triggered it is not marked as complete until the audit write succeeds. The system retries the write 3 times. If all retries fail, an alert is raised to QDB IT and the application is placed in a manual review hold.

US-23 — Document Encryption and Access Control

> As a QDB Data Protection Officer, I want uploaded documents to be encrypted at rest and accessible only to the applicant and authorized QDB staff so that PDPA obligations are met and commercially sensitive documents are protected.

Priority: P1 **Maps to**: BN-018, FR-006

Acceptance Criteria:

- **Given** a document is uploaded to the portal, **When** the file is written to storage, **Then** the file is encrypted at rest using AES-256. Encryption is applied before the file is written and is verified on read. The encryption key is managed via a key management service (not stored alongside the file).
- **Given** a document is in transit between the user's browser and the portal, **When** the upload occurs, **Then** the transport is secured with TLS 1.3 minimum. Any request over HTTP is rejected and redirected to HTTPS.
- **Given** an attempt to access a document via a direct storage URL, **When** the request is made without a valid portal session and document access token, **Then** the request is denied with HTTP 403. Document URLs are signed, time-limited (expiry: 1 hour), and scoped to the requesting user's case.
- **Given** an uploaded document passes virus scanning, **When** the scan completes, **Then** the document is moved to the active storage tier and the upload status is updated to "Validated". Documents that fail virus scanning are quarantined, the applicant is notified, and the event is logged.
- **Given** a document is requested by a QDB staff member via a CRM document link, **When** the request is made, **Then** the portal's document access service validates the staff member's role and generates a signed time-limited URL. The access event is recorded in the audit trail.

Bilingual Interface (BN-010)

US-24 — Bilingual Portal (Arabic RTL and English)

> As an Arabic-speaking SME owner, I want to use the portal entirely in Arabic so that language is not a barrier to accessing relief funding.

Priority: P1 **Maps to**: BN-010, FR-010

Acceptance Criteria:

- **Given** a user visits the portal home page, **When** they select their language preference (Arabic or English), **Then** all subsequent pages, labels, error messages, buttons, notifications, and system messages are rendered in the selected language.
- **Given** a user selects Arabic, **When** any page renders, **Then** the page layout is mirrored to RTL: text alignment is right, navigation flows right-to-left, and all Arabic text is typographically correct (no machine translation artifacts).
- **Given** a field contains data from an external system (e.g., MOCI returns company name in Arabic), **When** the field is displayed, **Then** the system-supplied Arabic text is displayed without modification. English equivalents are displayed alongside where MOCI provides both.
- **Given** a user's language preference is set to Arabic, **When** they receive an email or SMS notification, **Then** the notification content is in Arabic with correct RTL formatting in the email body.
- **Given** the language toggle is changed mid-session, **When** the new language is selected, **Then** the portal immediately re-renders the current page in the new language without losing form data or application progress.

Program Lifecycle (BN-015)

US-25 — Program Lifecycle Management (Open / Paused / Closed)

> As a QDB Program Administrator, I want the portal to enforce program lifecycle states so that applications are only accepted during the approved program window and the portal closes gracefully on the program end date.

Priority: P2 **Maps to**: BN-015, FR-012

Acceptance Criteria:

- **Given** the program end date is configured by a QDB Admin, **When** a user attempts to start a new application after the program end date, **Then** the portal displays: "The QDB SME Relief Program has closed. New applications are no longer accepted. If you have an existing application, you can track its status below." No new application session is created.
- **Given** an in-progress application was started before the program end date, **When** the applicant attempts to complete and submit it after the end date, **Then** the portal allows them to complete and submit. The grace period for in-progress applications is 48 hours after the program end date.
- **Given** a QDB Admin pauses the program (e.g., for system maintenance), **When** a user attempts to start or continue an application, **Then** the portal displays: "The portal is temporarily paused. Please try again shortly." Existing submitted applications are unaffected.
- **Given** the program is 14 days from its end date, **When** the notification threshold is reached, **Then** the QDB Compliance Officer and Program Admin receive an automated email alert: "The NRGP Relief Portal is scheduled to close on [DATE]. Review in-progress applications."

Duplicate Detection (BN-017)

US-26 — Detect and Reject Duplicate Applications

> As a system, I must detect and reject duplicate applications from the same CR number within the same relief period so that a single company cannot receive multiple disbursements.

Priority: P2 **Maps to**: BN-017, FR-003

Acceptance Criteria:

- **Given** a CR number has an active or completed application in the current relief period, **When** the same CR number is entered during a new application session, **Then** the portal displays: "An application for [Company Name] (CR: [NUMBER]) is already in progress. Your existing case reference is [CASE-ID]. You cannot submit a second application for the same relief period."
- **Given** the duplicate detection triggers, **When** the applicant views the duplicate notice, **Then** the portal provides a link to their existing application status page and a QDB Operations contact for exception handling.
- **Given** a duplicate is detected, **When** the event is logged, **Then** the audit trail records: detecting QID, CR number, existing case ID, detection timestamp.

4. Functional Requirements

FR-001 — NAS / Tawtheeq OAuth Integration

Attribute	Detail
ID	FR-001
User Stories	US-01, US-02
Description	The portal integrates with Qatar's National Authentication Service (NAS) via OIDC (preferred) or SAML 2.0 for user authentication. The portal does not store user credentials — it issues its own JWT session after receiving a validated NAS identity claim.
Inputs	OIDC authorization code; QID claim from NAS ID Token
Outputs	Portal JWT session (access token + refresh token); QID stored in session
Session Duration	30-minute inactivity timeout; maximum 8-hour absolute session limit
Error Handling	NAS unavailable → display fallback message with QDB Operations contact; auth failure → clear error with retry option
Dependency	QDB IT must provision NAS client credentials for this portal; reference QDB One ADR-002

FR-002 — MOCI CR Number Lookup Integration

Attribute	Detail
ID	FR-002
User Stories	US-03, US-04
Description	Given a CR number entered by the applicant, the portal queries the MOCI API to retrieve verified company data. The portal cross-references the authenticated QID against the returned signatory/shareholder list.
Inputs	CR number (10-digit); authenticated QID
Outputs	Company name (Arabic + English), CR status, sector, registration date, shareholder/signatory list
Latency	API response < 3 seconds
Error Handling	CR not found → display error; CR inactive/suspended → block application; MOCI unavailable → display error, do not allow manual bypass
Data Retention	Company data cached for current session only; not persisted post-submission

FR-003 — Eligibility Engine

Attribute	Detail
ID	FR-003
User Stories	US-05, US-06, US-07, US-26
Description	A rule-based engine that evaluates a company against seven eligibility criteria (EC-001 through EC-007). Rules are stored in a configuration table and are admin-editable without code deployment. Results include a per-criterion pass/fail with reason codes.
Criteria	EC-001: Active CR; EC-002: ≥12 months registered; EC-003: SME by QDB definition; EC-004: WPS enrollment; EC-005: Impacted sector or declared revenue decline; EC-006: No QDB loan in default; EC-007: No judicial dissolution
Inputs	MOCI company data; WPS employee count; QDB internal loan status flag; sector classification
Outputs	ELIGIBLE or INELIGIBLE; per-criterion result with reason code; criteria snapshot (stored in audit)
Admin Interface	Criteria parameters configurable via admin UI (FR-011); changes logged with admin identity
Duplicate Detection	Checks existing active applications for same CR number in same relief period before proceeding

FR-004 — NRGP List Matching

Attribute	Detail
ID	FR-004
User Stories	US-08, US-09, US-20
Description	The NRGP beneficiary list is stored in a PostgreSQL table loaded from a CSV uploaded by QDB Admin. On each eligible application, the CR number is looked up in the list. Exact match → auto-disbursement path. No match → manual review path.
Match Logic	Exact CR number match; fuzzy matching (Levenshtein distance ≤ 1) flagged as "possible match" for admin review
List Management	Admin uploads CSV; system validates format; lists are versioned; active list is the most recently confirmed upload
Latency	Lookup < 1 second
Audit	NRGP list version at time of lookup is recorded in the audit trail

FR-005 — Dynamics CRM Integration

Attribute	Detail
ID	FR-005
User Stories	US-08, US-09
Description	The portal creates and updates case records in Microsoft Dynamics 365 CRM via the Dynamics 365 Web API (REST/OData) using OAuth 2.0 with an Azure AD service principal.
Case Types	`auto_nrgp` (initial status: `pending_disbursement`); `manual_review` (initial status: `pending_review`)
Case Payload	Company name, CR number, QID, NRGP match result, eligibility snapshot, WPS result, document storage URLs, audit log reference
Status Sync	Portal polls CRM for status updates every 5 minutes for active applications; updates applicant-facing status
Retry Logic	Up to 3 retries with exponential back-off on CRM write failure; failure alert to QDB Operations after all retries fail
Authentication	Azure AD service principal; secret rotation every 90 days

FR-006 — Document Upload and Storage

Attribute	Detail
ID	FR-006
User Stories	US-10, US-11, US-12, US-13, US-14, US-23
Description	Applicants upload salary evidence, WPS files, rent evidence, CR copies, and business licenses. Each file is virus-scanned, validated, encrypted, and stored in secure object storage linked to the CRM case.
Accepted Formats	PDF, JPEG, PNG, TIFF (general documents); CSV (WPS file)
Size Limits	10 MB per file; 50 MB per application total
Storage Path	`{case_id}/{doc_type}/{timestamp}_{filename}`
Encryption	AES-256 at rest; TLS 1.3 in transit
Access Control	Signed, time-limited URLs (1-hour expiry); role-based access: applicant (own docs only), QDB staff (all docs by case), audit (read-only)
Retention	7-year minimum (BR-008)
Virus Scan	All files scanned before activation; infected files quarantined and applicant notified

FR-007 — WPS File Parser and Validator

Attribute	Detail
ID	FR-007
User Stories	US-11, US-15, US-16
Description	The portal parses uploaded WPS CSV files and cross-validates payroll data against the live WPS API. Discrepancies exceeding 10% trigger a manual review flag on the CRM case. WPS API failure results in a `wps_unavailable` flag (not a block).
WPS CSV Format	Ministry of Labor standard: columns for employee ID, month, payroll amount, payment date
Validation Rules	Data no older than 90 days (BR-010); declared amount within 10% of WPS figure (else flag)
Fallback	WPS API unavailable → use uploaded WPS file only; set `wps_api_fallback` flag on CRM case
Output	WPS validation result (pass/discrepancy/no_records/unavailable); discrepancy percentage; stored in CRM as a structured note

FR-008 — Audit Trail Service

Attribute	Detail
ID	FR-008
User Stories	US-22
Description	An append-only audit log capturing every material system event, decision, and actor action during an application lifecycle. Written transactionally — application steps are not marked complete until the audit write succeeds.
Event Types	auth_success, auth_failure, cr_lookup, eligibility_result, nrgp_check, crm_case_created, document_uploaded, document_virus_result, wps_validation, status_change, notification_sent, session_terminated, admin_rule_change, nrgp_list_upload, program_lifecycle_change
Record Fields	event_type, timestamp_utc, actor_qid (or "system"), session_id, application_id, input_summary, output_summary, data_source_id, criteria_snapshot (for eligibility events)
Tamper Prevention	Append-only table constraints; no UPDATE/DELETE permissions on audit table; access monitored by alerting
Export	Admin can export all audit records for a given application or date range as JSON or CSV

FR-009 — Notification Service

Attribute	Detail
ID	FR-009
User Stories	US-18
Description	The portal sends email and SMS notifications to applicants on key status changes. Notifications are sent in the applicant's language preference. Delivery failures trigger retry and, after 3 failures, a QDB Operations alert.
Trigger Events	Application submitted, eligibility result (eligible/ineligible), CRM case created, status changed to under_review, approved, rejected, disbursed
Email	Standard SMTP with HTML email template in Arabic and English; RTL layout for Arabic
SMS	SMS gateway integration; message < 160 characters; language-aware
Retry	3 retries with 10-minute intervals; failure after 3 attempts → QDB Operations alert

FR-010 — Bilingual UI (Arabic RTL + English)

Attribute	Detail
ID	FR-010
User Stories	US-24
Description	All portal UI text is available in Arabic and English. Language preference is stored in the user's session and profile. Arabic mode triggers full RTL layout mirroring. Translations are professionally reviewed — not machine-translated.
Coverage	All page labels, button text, error messages, validation messages, status labels, notifications, email templates
RTL Support	CSS logical properties; full layout mirroring; icon/form element directionality
Language Persistence	Language selection persists for the session; stored in user profile for return visits
Admin Interface	Admin UI is English-only in MVP; Arabic admin UI deferred

FR-011 — QDB Admin Dashboard

Attribute	Detail
ID	FR-011
User Stories	US-07, US-19, US-20, US-21
Description	A secured admin section of the portal accessible only to QDB staff with the admin role. Provides: application overview, KPI dashboard, NRGP list management, eligibility rule configuration, and program lifecycle controls.
Authentication	Separate admin login (QDB staff credentials via Azure AD or NAS); multi-factor authentication required
Application Management	Table with filter/search; click-through to full application detail; export to CSV
KPI Dashboard	Total applications, status counts, average processing time, WPS discrepancy rate, document re-submission rate
NRGP List	Upload/validate/activate new list; view active list details; download current list
Eligibility Rules	View/edit rule parameters; rule change history
Program Lifecycle	Set program status (open/paused/closed) and end date

FR-012 — Program Lifecycle Management

Attribute	Detail
ID	FR-012
User Stories	US-25
Description	The portal enforces program lifecycle states: Open (accepting applications), Paused (temporarily suspended), Closed (no new applications; read-only for status tracking). Lifecycle is managed by QDB Admin via FR-011.
Lifecycle States	OPEN, PAUSED, CLOSED
Program End Date	Configured by admin; hard-coded enforcement at midnight on end date
Grace Period	48-hour grace for in-progress applications started before end date
Notifications	14-day and 7-day pre-close alerts to QDB Admin and Compliance Officer
Post-Close Behavior	Portal home page shows closure message; status tracking remains available

5. Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-001	The portal must support 500 concurrent applicant sessions without response time degradation.
NFR-002	All page load times must be < 2 seconds under normal load (100 concurrent users).
NFR-003	MOCI CR lookup response must complete within 3 seconds (end-to-end, including API roundtrip).
NFR-004	The eligibility engine must return a result within 5 seconds of initiating evaluation.
NFR-005	CRM case creation must complete and return a Case ID within 5 seconds.
NFR-006	The portal must handle a peak submission spike of 200 simultaneous sessions (program launch surge) without error rate exceeding 0.5%.

5.2 Security

ID	Requirement
NFR-007	All authentication must be via NAS/Tawtheeq OIDC; the portal stores no user passwords.
NFR-008	All data in transit must be secured with TLS 1.3 minimum. HTTP requests must be rejected and redirected to HTTPS.
NFR-009	Documents must be encrypted at rest using AES-256 with keys managed by a dedicated key management service.
NFR-010	Document access URLs must be signed and time-limited (maximum 1-hour expiry).
NFR-011	The portal must comply with Qatar's Personal Data Protection Act (PDPA). A PDPA impact assessment must be completed and signed off before launch.
NFR-012	All uploaded documents must be virus-scanned before being moved to active storage.
NFR-013	Admin access must require multi-factor authentication.
NFR-014	Rate limiting must be applied per CR number (max 3 application attempts per CR per 24 hours).
NFR-015	The audit trail must be append-only. Database-level constraints must prevent UPDATE and DELETE on audit records.
NFR-016	Document retention must be enforced for a minimum of 7 years in accordance with QDB policy (BR-008).

5.3 Availability and Reliability

ID	Requirement
NFR-017	Portal availability must be ≥ 99.5% during the program window (business hours, Sunday–Thursday, 08:00–18:00 AST, Qatar timezone).
NFR-018	The portal must gracefully degrade when external APIs (MOCI, WPS, CRM) are unavailable — displaying informative error messages and, where applicable, flagging for manual follow-up rather than returning unhandled errors.
NFR-019	CRM case creation failures must be queued for retry; the portal must not lose application data due to transient CRM unavailability.
NFR-020	Audit log write failures must block application step completion. The system must not silently skip audit events.

5.4 Accessibility

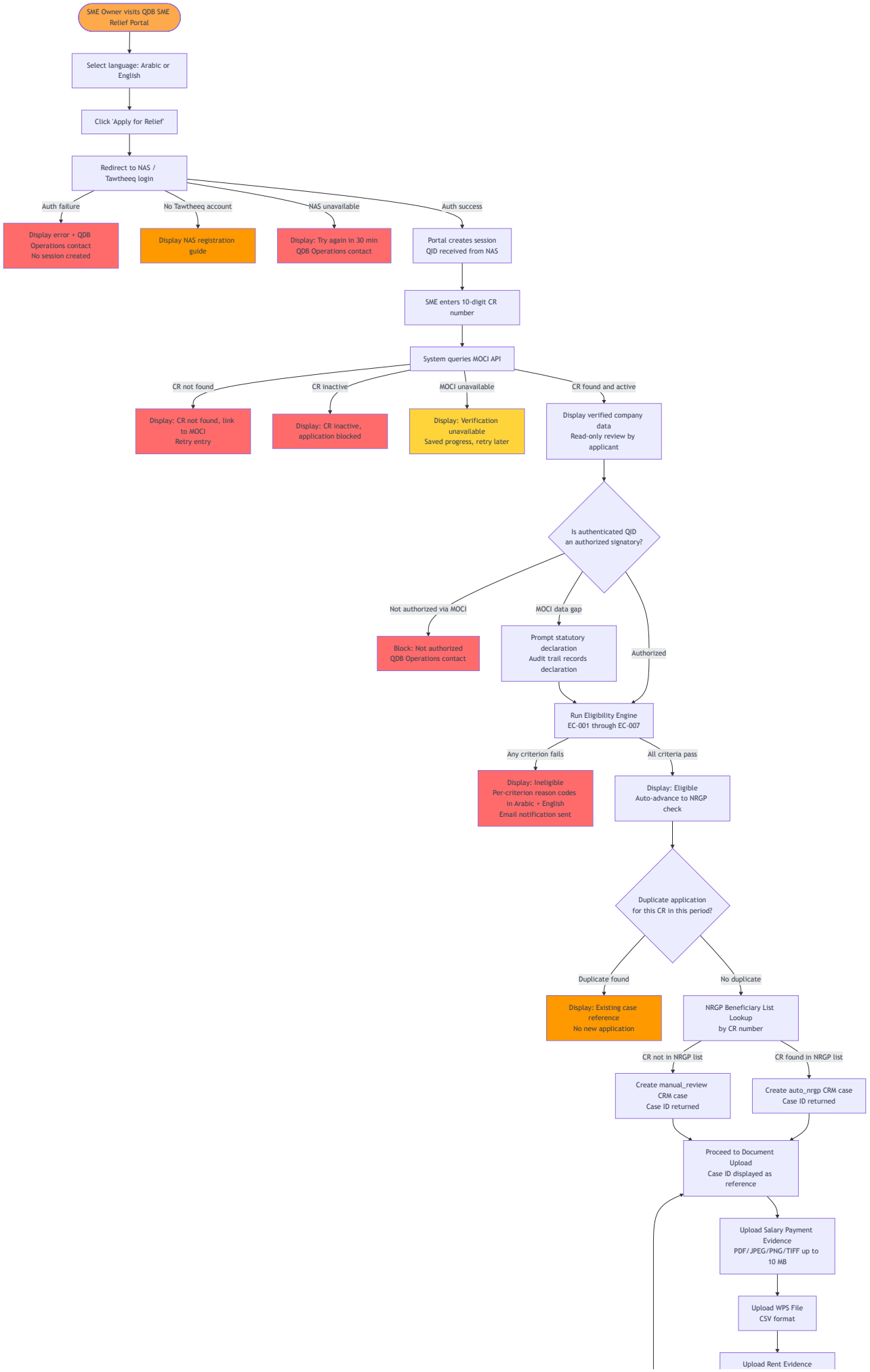
ID	Requirement
NFR-021	The portal must meet WCAG 2.1 AA compliance for all applicant-facing pages.
NFR-022	All form fields must have accessible labels. All error messages must be announced by screen readers.
NFR-023	The Arabic RTL layout must be fully tested and functional. Arabic text must be professionally translated (not machine-translated).
NFR-024	The portal must be responsive and usable on desktop, tablet, and mobile viewports.

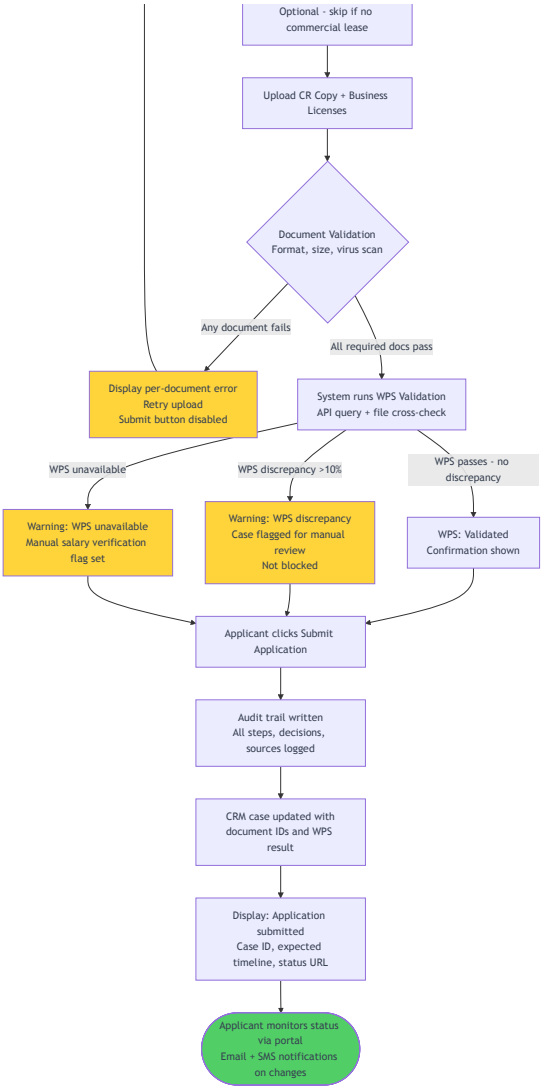
5.5 Maintainability

ID	Requirement
NFR-025	Eligibility criteria must be admin-configurable without code deployment (FR-003, FR-011).
NFR-026	The NRGP beneficiary list must be updatable via admin upload without engineer intervention (FR-004, FR-011).
NFR-027	Program lifecycle (open/paused/closed) must be controllable via admin UI without code deployment (FR-012).

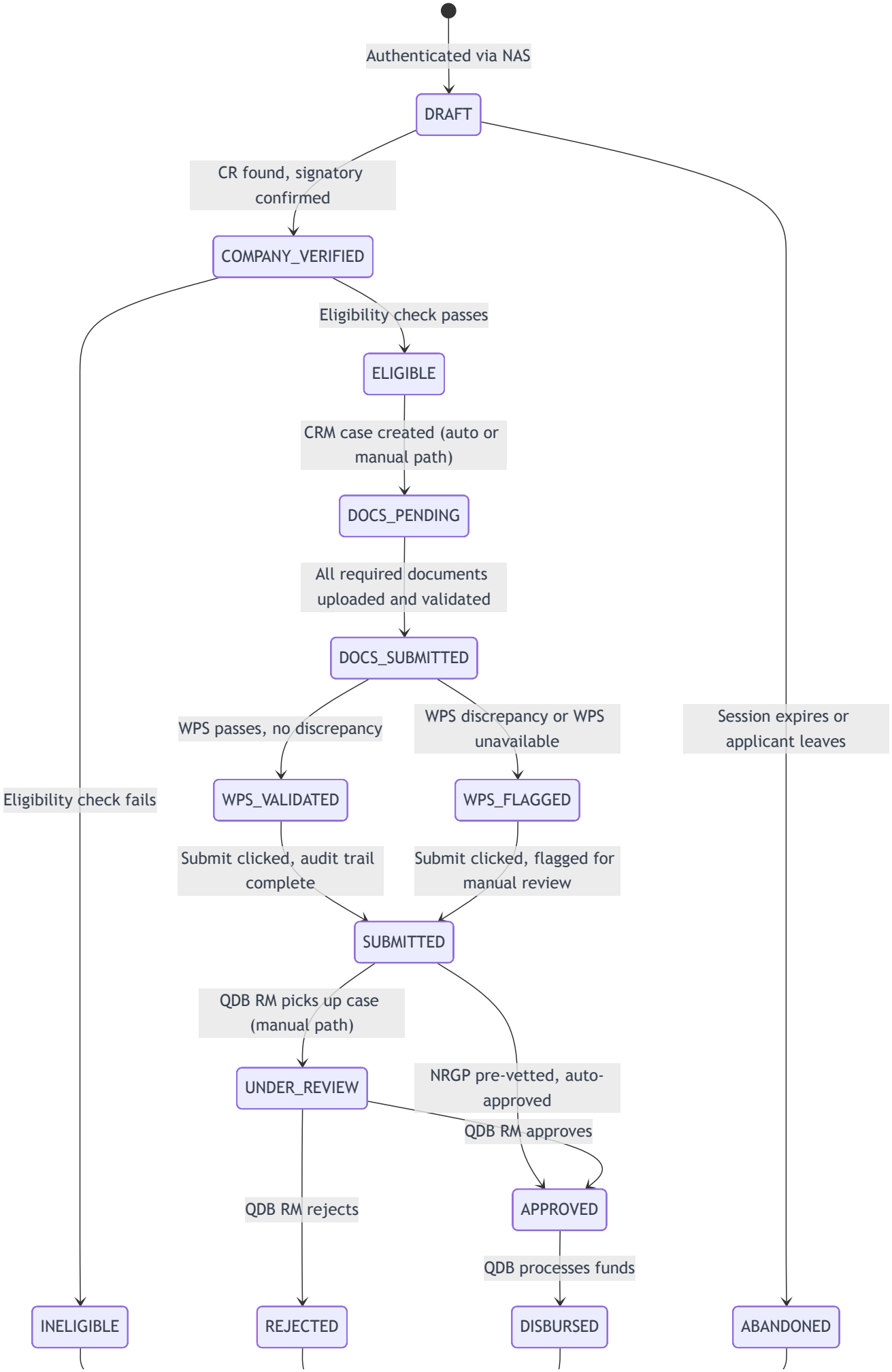
6. Mermaid Diagrams

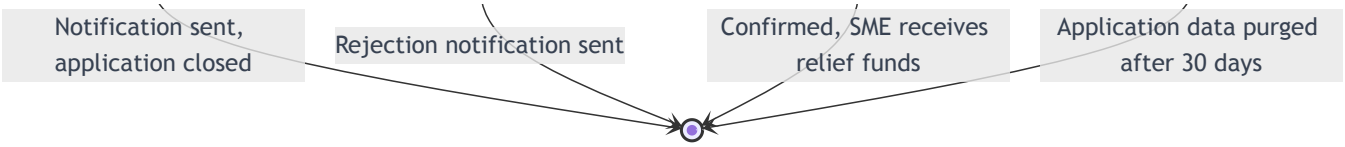
6.1 Full End-to-End Applicant Journey Flowchart


SME Owner visits QDB SME Relief Portal
Select language: Arabic or English
Click 'Apply for Relief'
Redirect to NAS / Tawtheeq login
Auth failure
Display error + QDB Operations contact No session created
No Tawtheeq account
Display NAS registration guide
NAS unavailable
Display: Try again in 30 min QDB Operations contact
Auth success
Portal creates session QID received from NAS
SME enters 10-digit CR number
System queries MOCI API
CR not found
Display: CR not found, link to MOCI Retry entry
CR inactive
Display: CR inactive, application blocked
MOCI unavailable
Display: Verification unavailable Saved progress, retry later
CR found and active
Display verified company data Read-only review by applicant
Is authenticated QID an authorized signatory?
Not authorized via MOCI
Block: Not authorized QDB Operations contact
MOCI data gap
Prompt statutory declaration Audit trail records declaration
Authorized
Run Eligibility Engine EC-001 through EC-007
Any criterion fails
Display: Ineligible Per-criterion reason codes in Arabic + English Email notification sent
All criteria pass
Display: Eligible Auto-advance to NRGP check
Duplicate application for this CR in this period?
Duplicate found
Display: Existing case reference No new application
No duplicate
NRGP Beneficiary List Lookup by CR number
CR not in NRGP list
Create manual_review CRM case Case ID returned
CR found in NRGP list
Create auto_nrgp CRM case Case ID returned
Proceed to Document Upload Case ID displayed as reference
Upload Salary Payment Evidence PDF/JPEG/PNG/TIFF up to 10 MB
Upload WPS File CSV format
Upload Rent Evidence

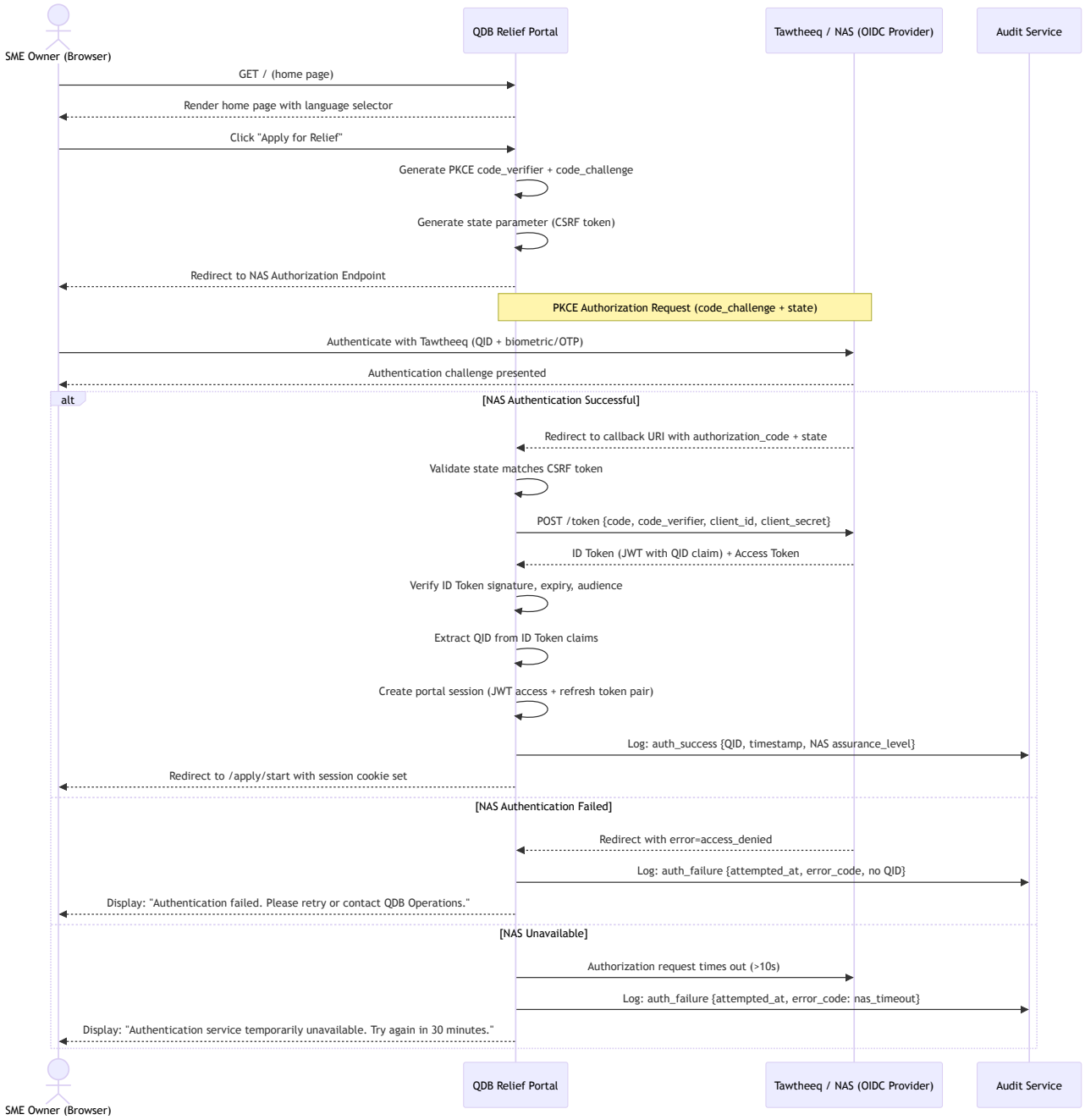


6.2 Application State Diagram

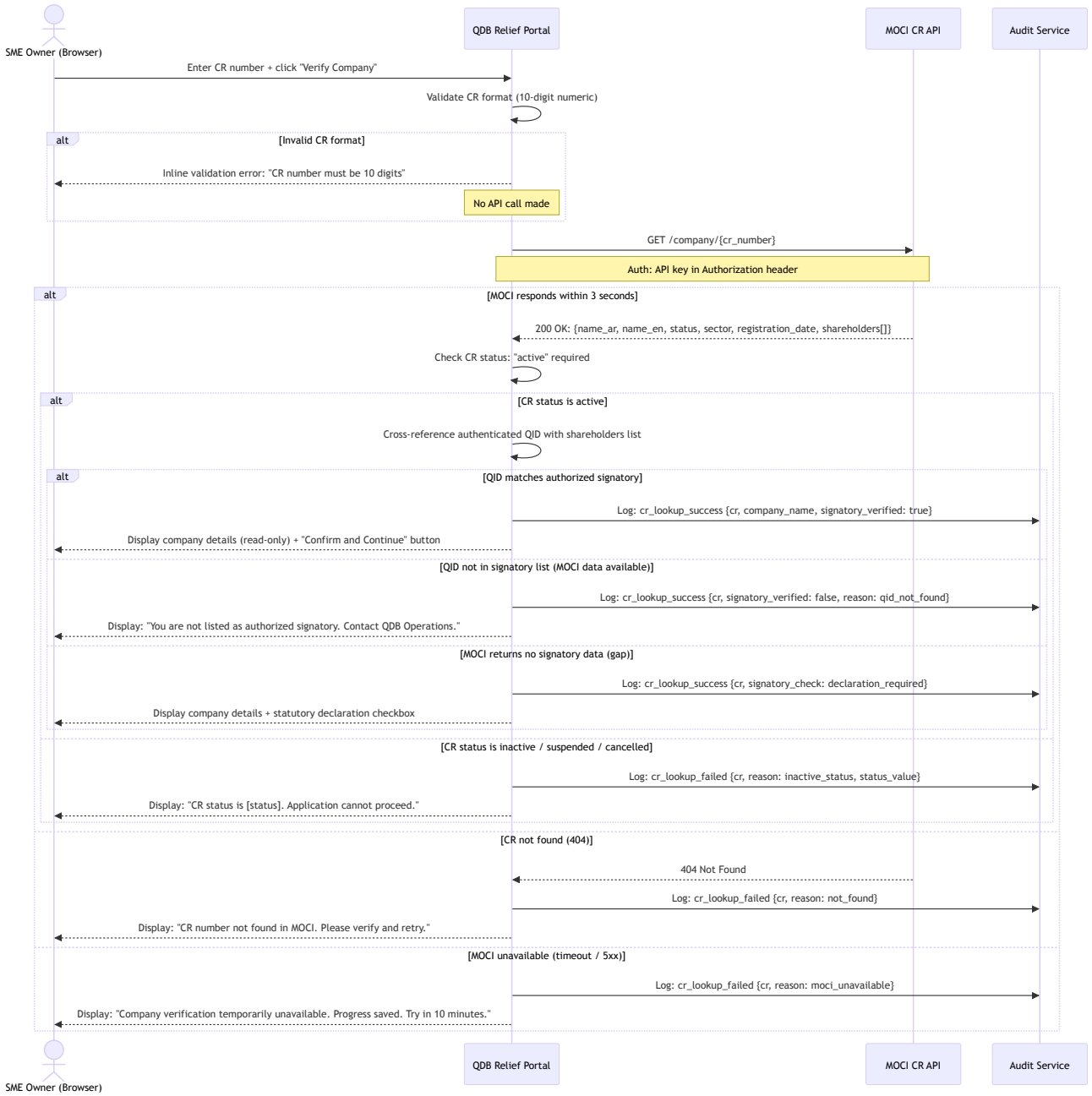

Authenticated via NAS
DRAFT
CR found, signatory confirmed
COMPANY_VERIFIED
Eligibility check passes
ELIGIBLE
CRM case created (auto or manual path)
DOCS_PENDING
All required documents uploaded and validated
DOCS_SUBMITTED
WPS passes, no discrepancy
WPS discrepancy or WPS unavailable
Session expires or applicant leaves
Eligibility check fails
WPS_VALIDATED
WPS_FLAGGED
Submit clicked, audit trail complete
Submit clicked, flagged for manual review
SUBMITTED
QDB RM picks up case (manual path)
UNDER_REVIEW
NRGP pre-vetted, auto-approved
QDB RM approves
QDB RM rejects
APPROVED
QDB processes funds
INELIGIBLE
REJECTED
DISBURSED
ABANDONED



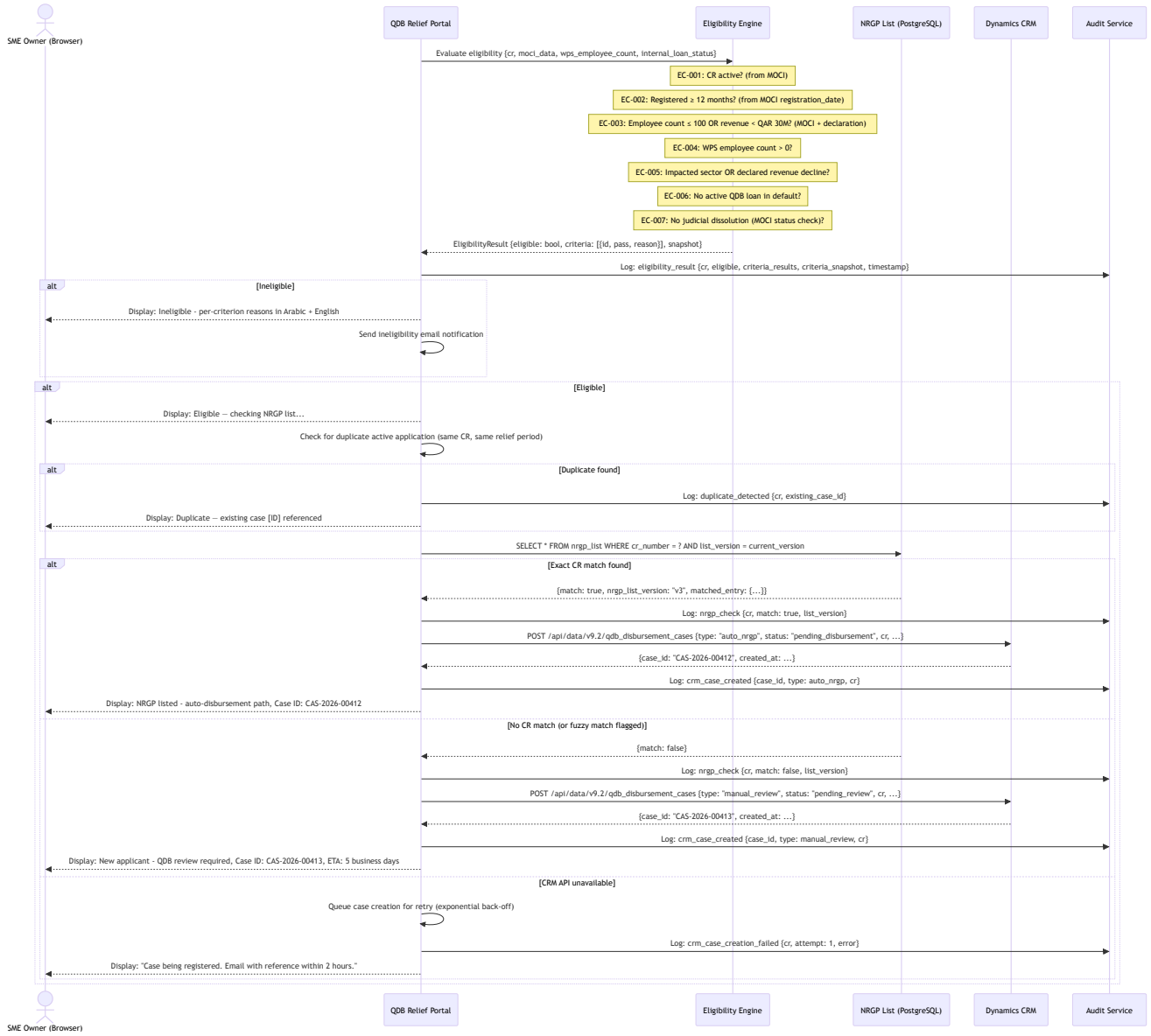
6.3 Tawtheeq / NAS Authentication Sequence



6.4 MOCI CR Number Lookup Sequence

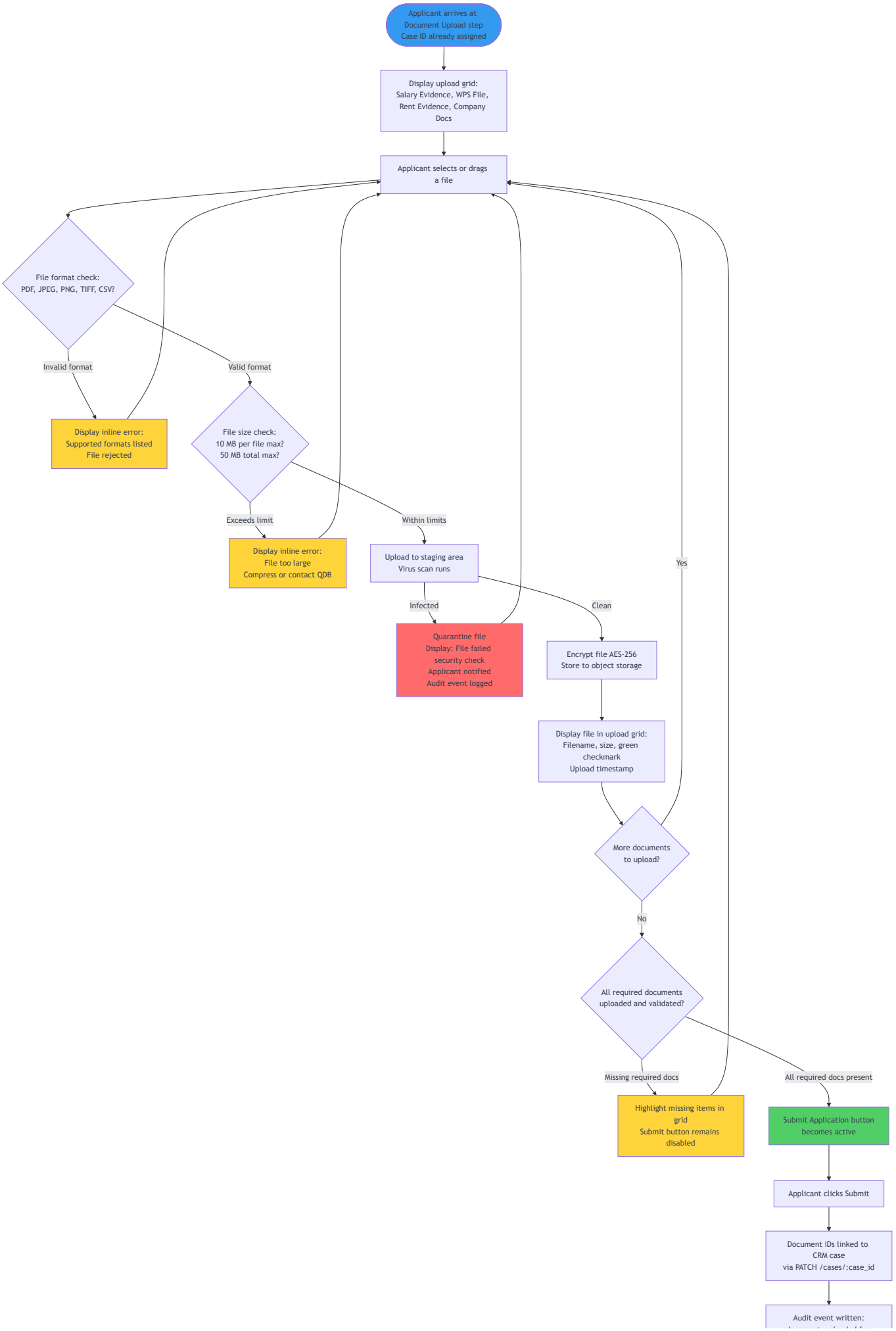


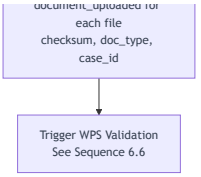
6.5 Eligibility Engine + NRGP Check + CRM Routing Sequence



6.6 WPS Validation Sequence

6.7 Document Upload Flow


Applicant arrives at Document Upload step Case ID already assigned
Display upload grid: Salary Evidence, WPS File, Rent Evidence, Company Docs
Applicant selects or drags a file
File format check: PDF, JPEG, PNG, TIFF, CSV?
Invalid format
Display inline error: Supported formats listed File rejected
Valid format
File size check: 10 MB per file max? 50 MB total max?
Exceeds limit
Display inline error: File too large Compress or contact QDB
Within limits
Upload to staging area Virus scan runs
Infected
Quarantine file Display: File failed security check Applicant notified Audit event logged
Clean
Encrypt file AES-256 Store to object storage
Display file in upload grid: Filename, size, green checkmark Upload timestamp
More documents to upload?
Yes
No
All required documents uploaded and validated?
Missing required docs
Highlight missing items in grid Submit button remains disabled
All required docs present
Submit Application button becomes active
Applicant clicks Submit
Document IDs linked to CRM case via PATCH /cases/:case_id
Audit event written:



7. Integration Specifications

7.1 NAS / Tawtheeq (OIDC)

Protocol: OpenID Connect Authorization Code Flow with PKCE

Auth Mechanism: Client ID + Client Secret (server-side); PKCE for CSRF protection

Authorization Endpoint Request:

```
GET https://nas.gov.qa/auth/realms/tawtheeq/protocol/openid-connect/auth
  ?client_id=qdb-relief-portal
  &response_type=code
  &scope=openid qid profile
  &redirect_uri=https://relief.qdb.com.qa/auth/callback
  &state={csrf_token}
  &code_challenge={pkce_challenge}
  &code_challenge_method=S256
```

Token Request:

```
POST https://nas.gov.qa/auth/realms/tawtheeq/protocol/openid-connect/token
Content-Type: application/x-www-form-urlencoded

grant_type=authorization_code
&code={authorization_code}
&redirect_uri=https://relief.qdb.com.qa/auth/callback
&client_id=qdb-relief-portal
&client_secret={secret}
&code_verifier={pkce_verifier}
```

ID Token Claims (expected):

```
{
  "sub": "qas-user-uuid",
  "qid": "28712345678",
  "name": "Khalid Al-Mansouri",
  "name_ar": "خالد المنصوري",
  "assurance_level": 2,
  "iat": 1740000000,
  "exp": 1740003600
}
```

Error Handling:

Scenario	Behaviour
Auth failed (user error)	Redirect back with `error=access_denied`; portal shows retry message
NAS timeout (>10s)	Display "Service temporarily unavailable" with QDB Operations contact
Invalid state (CSRF)	Session terminated; user redirected to start

Fallback: Display QDB Operations contact information. No alternative authentication path in MVP.

7.2 MOCI CR Number Lookup (REST API)

Protocol: REST (JSON) — confirm SOAP availability with MOCI if REST is not available.

Auth Mechanism: API Key in `Authorization: Bearer {api_key}` header.

Request:

```
GET https://api.moci.gov.qa/v1/companies/{cr_number}
Authorization: Bearer {api_key}
Accept: application/json
```

Success Response (200 OK):

```
{
  "cr_number": "1234567890",
  "name_en": "Al Mansouri Trading LLC",
  "name_ar": "شركة المنصوري للتجارة ذ.م.م",
  "status": "active",
  "registration_date": "2021-03-15",
  "sector": "wholesale_trade",
  "employee_count": 42,
  "shareholders": [
    {
      "qid": "28712345678",
      "name_en": "Khalid Al-Mansouri",
      "name_ar": "خالد المنصوري",
      "role": "authorized_signatory"
    }
  ]
}
```

Error Responses:

HTTP Status	Meaning	Portal Behaviour
404	CR not found	Display: "CR not found" with MOCI link
200 with `status: inactive`	CR inactive	Display: "CR is inactive" — block application
503 / timeout	MOCI unavailable	Display: "Temporarily unavailable" — save progress

Fallback: Do not allow manual data entry. Portal does not proceed until MOCI responds.

7.3 WPS System (Salary Validation)

Protocol: REST API (preferred) or secure file transfer — confirm with Ministry of Labor.

Auth Mechanism: MOU-secured service account credentials; OAuth 2.0 or mTLS.

Request:

```
GET https://wps.mol.gov.qa/v1/payroll/{cr_number}?period=last_90_days
Authorization: Bearer {service_token}
```

Success Response (200 OK):

```
{
  "cr_number": "1234567890",
  "query_date": "2026-03-03",
  "employee_count": 42,
  "total_payroll_90d": 1050000.00,
  "currency": "QAR",
  "monthly_records": [
    {"month": "2025-12", "total_amount": 350000.00, "payment_date": "2025-12-28", "employees
    {"month": "2026-01", "total_amount": 350000.00, "payment_date": "2026-01-27", "employees
    {"month": "2026-02", "total_amount": 350000.00, "payment_date": "2026-02-26", "employees
  ]
}
```

Error Handling:

Scenario	Behaviour
200 with empty records	Set `wps_no_records` flag; manual verification required
503 / timeout	Set `wps_api_fallback` flag; fall back to uploaded WPS CSV
MOU not in place	Full fallback to document-based salary verification; WPS API not called

---

7.4 Microsoft Dynamics CRM (Disbursement Case Management)

Protocol: Dynamics 365 Web API (REST / OData v4.0)

Auth Mechanism: OAuth 2.0 Client Credentials (Azure AD service principal)

Token Request:

```
POST https://login.microsoftonline.com/{tenant_id}/oauth2/v2.0/token
Content-Type: application/x-www-form-urlencoded

grant_type=client_credentials
&client_id={sp_client_id}
&client_secret={sp_secret}
&scope=https://{crm_org}.crm.dynamics.com/.default
```

Create Case Request:

```
POST https://{crm_org}.crm.dynamics.com/api/data/v9.2/qdb_disbursement_cases
Authorization: Bearer {access_token}
Content-Type: application/json
OData-Version: 4.0

{
  "qdb_casetypecode": "auto_nrgp",
  "statuscode": 1,
  "qdb_crnumber": "1234567890",
  "qdb_companyname_en": "Al Mansouri Trading LLC",
  "qdb_companyname_ar": "شركة المنصوري للتجارة ذ.م.م",
  "qdb_applicantqid": "28712345678",
  "qdb_nrgplistmatch": true,
  "qdb_nrgplistversion": "v3",
  "qdb_eligibilityresult": "eligible",
  "qdb_submissiontimestamp": "2026-03-03T10:45:00Z",
  "qdb_portalapplicationid": "APP-2026-00501",
  "qdb_documenturls": "[]",
  "qdb_wpsvalidationresult": "pending"
}
```

Success Response:

```
{
  "@odata.context": "...",
  "qdb_disbursementcaseid": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "qdb_casenumber": "CAS-2026-00412"
}
```

Update Case (after document upload):

```
PATCH https://{crm_org}.crm.dynamics.com/api/data/v9.2/qdb_disbursement_cases({case_guid})
Authorization: Bearer {access_token}
Content-Type: application/json

{
  "qdb_documenturls": "[\"https://docs.qdb.internal/APP-2026-00501/salary_evidence/...\"]",
  "qdb_wpsvalidationresult": "pass",
  "qdb_applicationstatus": "submitted"
}
```

Error Handling:

Scenario	Behaviour
503 / timeout	Retry 3x with exponential back-off; queue for async creation; alert QDB Ops
401 Unauthorized	Refresh access token; retry; alert QDB IT if token refresh fails
400 Bad Request	Log error with payload; alert QDB IT; flag application for manual case creation

---

7.5 Document Storage

Service: Azure Blob Storage (Qatar-resident region) or equivalent on-premise object storage.

Auth Mechanism: Managed Identity or service account with least-privilege read/write per container.

Upload Flow:

```
1. Portal receives file from client (multipart/form-data)
2. Virus scan via integrated AV service
3. If clean: encrypt (AES-256) and write to storage
   Path: {container}/{case_id}/{doc_type}/{timestamp}_{sanitized_filename}
4. Return: {document_id, storage_path, checksum_sha256}
5. Link document to CRM case via PATCH (FR-005)
```

Download (Signed URL):

```
GET /api/documents/{document_id}/url
Authorization: Bearer {portal_session_token}

Response:
{
  "signed_url": "https://storage.qdb.internal/...?sig={signature}&se={expiry}",
  "expires_at": "2026-03-03T11:45:00Z"
}
```

Access Tiers:

Actor	Access Level
Applicant	Upload to own case; read own documents (signed URL, 1h)
QDB Relationship Manager	Read all documents for assigned cases (signed URL via CRM link)
QDB Compliance / Audit	Read all documents (audit export function)
System	Write (upload), read (for CRM linking)

8. Acceptance Criteria Summary

US-XX	Title	Given	When	Then	Priority
US-01	Tawtheeq Login	User visits portal and clicks Apply	Completes NAS authentication	Portal creates session, user reaches /apply/start	P0
US-02	Session Mgmt	User is authenticated	User idles 30 minutes	Warning shown; session expires; progress saved	P0
US-03	MOCI CR Lookup	Authenticated user enters valid CR	Verify Company clicked	MOCI data displayed read-only within 3 seconds	P0
US-04	Signatory Check	CR data returned	QID cross-referenced with MOCI	Match: proceed; Mismatch: block; Gap: declaration captured	P0
US-05	Eligibility Engine	Signatory confirmed	Engine runs	All 7 criteria evaluated; result in < 5 seconds	P0
US-06	Eligibility Result	Ineligible result returned	Result page renders	Per-criterion reasons in Arabic + English shown	P0
US-07	Admin Eligibility Config	Admin authenticated	Admin changes criterion value	Change takes effect immediately; logged in audit	P2
US-08	Auto CRM Case	CR found in NRGP list	NRGP check completes	auto_nrgp CRM case created; Case ID returned within 5s	P0
US-09	Manual CRM Case	CR not in NRGP list	NRGP check completes	manual_review CRM case created; applicant notified	P0
US-10	Salary Upload	On document upload step	File uploaded	Accepted formats/sizes validated; stored encrypted	P0
US-11	WPS File Upload	On document upload step	WPS CSV uploaded	File parsed; summary displayed; cross-checked with WPS API	P0
US-12	Rent Upload	On document upload step	Rent evidence uploaded	Files stored; optional if no commercial lease	P0

US-XX	Title	Given	When	Then	Priority
US-13	Company Docs Upload	On document upload step	CR copy + licenses uploaded	Up to 5 licenses accepted; linked to CRM case	P0
US-14	Upload Status	Documents uploaded	Applicant views upload section	Per-file status indicators; Submit enabled only when required docs pass	P0
US-15	WPS Validation	WPS API query initiated	WPS responds	Within-10% match: pass; >10%: flag on CRM; no records: manual flag	P0
US-16	WPS Result View	WPS validation complete	Applicant views result	Comparison shown; warning (non-blocking) if discrepancy	P1
US-17	Application Status	Application submitted	User views status page	Current status; history log; next step; in Arabic or English	P1
US-18	Notifications	Status changes in system	Event fires	Email + SMS sent within 5 minutes; language-aware	P1
US-19	QDB Dashboard	QDB Admin authenticated	Views Applications panel	Table with filters; KPI summary; click-through to detail	P1
US-20	NRGP List Upload	Admin on NRGP List page	Uploads new CSV	Validated; previewed; activated after confirm; old list deactivated	P1
US-21	Admin Eligibility Config	Admin authenticated	Views Eligibility Rules	See US-07	P2
US-22	Audit Trail	Any application step occurs	Step completes	Audit record written; append-only; audit write failure blocks step	P0
US-23	Document Encryption	Document uploaded	File written to storage	AES-256 at rest; TLS 1.3 in transit; signed URL with 1h expiry	P1

US-XX	Title	Given	When	Then	Priority
US-24	Bilingual UI	User selects Arabic	Page renders	Full Arabic RTL layout; professional translation; notifications in Arabic	P1
US-25	Program Lifecycle	Program end date reached	User attempts new application	Application blocked; closure message shown; in-progress apps get 48h grace	P2
US-26	Duplicate Detection	CR has existing active application	Same CR entered in new session	Duplicate blocked; existing case ID shown	P2

9. Site Map

All routes must exist as real pages. Status "MVP" routes are fully implemented. Status "Deferred" routes are real pages with a skeleton and empty state — never 404 errors.

Route	Status	Description	Auth Required
`/`	MVP	Portal home: language selector, Apply button, program overview	No
`/auth/login`	MVP	Initiates NAS OIDC flow; handles callback	No
`/auth/callback`	MVP	NAS redirect target; processes code; creates session	No
`/auth/logout`	MVP	Invalidates session; redirects to home	Yes
`/apply/start`	MVP	Application start page; duplicate check; CR entry form	Yes
`/apply/company`	MVP	MOCI company data review; signatory confirmation	Yes
`/apply/eligibility`	MVP	Eligibility check progress and result	Yes
`/apply/documents`	MVP	Document upload grid (salary, WPS, rent, company docs)	Yes
`/apply/review`	MVP	Application summary before submission	Yes
`/apply/confirmation`	MVP	Submission confirmation; Case ID; status link	Yes
`/application/status`	MVP	Applicant status page; history log; CRM status sync	Yes
`/admin`	MVP	QDB Admin dashboard home; redirects to /admin/applications	Admin only
`/admin/applications`	MVP	Application overview table with filters and KPIs	Admin only
`/admin/applications/:id`	MVP	Individual application detail view	Admin only
`/admin/nrgp-list`	MVP	NRGP beneficiary list management (upload, activate, view)	Admin only
`/admin/eligibility-rules`	MVP	Eligibility criteria configuration	Admin only
`/admin/program-lifecycle`	MVP	Program status control (open/paused/closed) and end date	Admin only

Route	Status	Description	Auth Required
`/help`	MVP	Help/FAQ page; program overview; contact details	No
`/program-closed`	MVP	Displayed when program is closed; status tracking link remains	No

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10. Out of Scope

The following are explicitly excluded from this portal:

1. **Payment processing and fund transfer**: The disbursement of funds is handled by QDB's internal banking systems after CRM case approval. The portal creates and routes cases but does not initiate or track fund transfers.
2. **QDB CRM UI**: The portal does not replace or replicate the Dynamics CRM interface. QDB Relationship Managers work in CRM directly; the portal creates records and attaches documents.
3. **Mobile native application**: The portal is a web application, responsive for mobile browsers. No native iOS or Android application is in scope for this release.
4. **Appeal workflow**: The portal surfaces rejection reasons but does not manage a formal appeal process. Appeals are handled offline via QDB Operations.
5. **Loan servicing**: The portal covers the application intake phase only. Loan agreement signing, repayment scheduling, and ongoing loan management are out of scope.
6. **SME onboarding to Tawtheeq**: The portal provides guidance and a QDB contact for SMEs without Tawtheeq accounts but does not facilitate the NAS registration process itself.
7. **Integration with QDB's core banking system**: The portal integrates with Dynamics CRM only. Core banking integration (for fund disbursement) is QDB's internal responsibility.
8. **Analytics and BI reporting beyond the admin dashboard**: Detailed reporting, Excel exports beyond the admin dashboard, or Power BI integration is deferred.
9. **Multi-period applications**: An SME may only apply once per relief period. If QDB extends the program to cover multiple distinct periods, multi-period logic is a new feature.
10. **Arabic admin interface**: The admin dashboard is English-only in MVP. An Arabic admin interface is deferred.

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11. Open Questions and Assumptions Requiring Resolution

These items are blockers or high-risk dependencies. They must be resolved before or during Sprint 0. The source is the BA Risk Register (RSK-XXX) and Assumptions list (ASM-XXX).

BLOCKER — Must Resolve Before Sprint 1 Starts

ID	Question	Risk if Unresolved	Owner	Target
OQ-001	Is the WPS API available for integration? What is the protocol (REST / file transfer / SOAP)? Is a data sharing MOU between QDB and Ministry of Labor already in place?	WPS validation cannot be automated. Salary relief amounts rely on applicant declarations only — significant fraud risk (RSK-001)	Architect + QDB Legal	Sprint 0
OQ-002	Can QDB obtain NAS sandbox credentials for this portal's development environment? What is the timeline for production onboarding?	Auth integration cannot be tested; development must use mock OIDC; production launch may be delayed (RSK-002)	Backend Engineer + QDB IT	Sprint 0
OQ-003	Have the NRGP eligibility criteria been formally documented and signed off by QDB Credit Risk and Compliance? Specifically: SME employee threshold, revenue threshold, impacted sector list, and revenue decline percentage for EC-005	Eligibility engine cannot be coded until criteria are signed off. Risk of rework mid-sprint (RSK-007)	Product Manager + QDB Credit Risk	Sprint 0 exit gate
OQ-004	What is the format of the current NRGP beneficiary list (CSV, Excel, database export)? How many records? Is it clean and deduped?	Data Engineer needs to assess cleansing effort; NRGP list feature may be delayed (RSK-006)	Data Engineer + QDB Operations	Sprint 0
OQ-005	Does the MOCI API return shareholder / authorized signatory data? If not, the portal falls back to a statutory declaration — what is QDB's acceptance of this fallback for legal enforceability?	Authorization check is the key fraud control. If MOCI does not provide signatory data, the check is weakened (RSK-003)	Architect + QDB Legal	Sprint 0

HIGH PRIORITY — Resolve Before Sprint 2 Starts

ID	Question	Risk if Unresolved	Owner	Target
OQ-006	What Dynamics CRM entities and schema are available for this portal? Does the CRM schema require customisation (new entities, custom fields)? What is QDB IT's change management timeline for CRM schema changes?	CRM integration may be delayed by IT change management (RSK-004)	DevOps + QDB IT	Sprint 1
OQ-007	What percentage of the target SME population already has Tawtheeq accounts? If > 20% lack accounts, an alternative authentication path or assisted onboarding must be designed — expanding scope	A significant authentication gap would mean a significant portion of eligible SMEs cannot apply (RSK-005)	Product Manager	Sprint 0 (survey)
OQ-008	What are the specific relief quantum caps? Salary relief cap per employee per month, rent relief cap per month, and combined total cap per application? These values must be hard-coded or admin-configurable before launch.	Portal cannot display expected relief amounts to applicants; CRM cases will have incomplete quantum information	Product Manager + QDB Credit Risk	Sprint 1
OQ-009	What is the confirmed program start date and end date? Is the program window fixed or subject to extension by QDB board decision?	Portal lifecycle management (FR-012) cannot be configured; program end date enforcement cannot be tested	Product Manager + QDB Program Office	Sprint 0

MEDIUM PRIORITY — Resolve Before Launch

ID	Question	Risk if Unresolved	Owner	Target
OQ-010	Has a PDPA compliance impact assessment been initiated? Who is the QDB Data Protection Officer responsible for sign-off before launch?	PDPA non-compliance is a legal risk for QDB (RSK-012)	Security Engineer + QDB Legal	Sprint 3
OQ-011	What is the Arabic translation process? Will QDB provide professional legal/financial Arabic translations for all portal copy, or is ConnectSW responsible for sourcing a translation service?	Poor Arabic translation quality undermines user trust and accessibility (RSK-011)	UI/UX Designer + QDB Comms	Sprint 2
OQ-012	What are the specific sector codes classified as "impacted" for EC-005? Which MOCI sector classifications qualify as having demonstrable geopolitical exposure?	Eligibility criterion EC-005 cannot be implemented in the automated engine without the sector list	Product Manager + QDB Credit Risk	Sprint 1
OQ-013	Is the QDB internal loan default status (EC-006) available via an API or must it be a manual QDB flag set by Operations before NRGP list processing?	EC-006 may require a manual pre-screening step rather than automated evaluation, affecting architecture	Architect + QDB IT	Sprint 1

12. Traceability Matrix

US-XX	Story Title	BN-XXX (Business Need)	FR-XXX (Functional Req)	Priority
US-01	Tawtheeq Login	BN-001	FR-001	P0
US-02	Session Management	BN-001	FR-001	P0
US-03	MOCI CR Lookup	BN-002	FR-002	P0
US-04	Signatory Check	BN-002, BN-008	FR-002	P0
US-05	Eligibility Engine	BN-003	FR-003	P0
US-06	Eligibility Result Display	BN-003, BN-009	FR-003	P0
US-07	Admin Eligibility Config	BN-016	FR-003, FR-011	P2
US-08	Auto CRM Case (NRGP match)	BN-004	FR-004, FR-005	P0
US-09	Manual CRM Case (no NRGP match)	BN-005	FR-004, FR-005	P0
US-10	Salary Evidence Upload	BN-006	FR-006	P0
US-11	WPS File Upload	BN-007	FR-006, FR-007	P0
US-12	Rent Evidence Upload	BN-006	FR-006	P0
US-13	Company Document Upload	BN-006	FR-006	P0
US-14	Upload Status Display	BN-006, BN-011	FR-006	P0
US-15	WPS Validation (system)	BN-007	FR-007	P0
US-16	WPS Result View (applicant)	BN-007, BN-009	FR-007	P1
US-17	Application Status View	BN-009	FR-008, FR-005	P1
US-18	Status Change Notifications	BN-012	FR-009	P1
US-19	QDB Operations Dashboard	BN-013	FR-011	P1

US-XX	Story Title	BN-XXX (Business Need)	FR-XXX (Functional Req)	Priority
US-20	NRGP List Upload (Admin)	BN-013	FR-004, FR-011	P1
US-21	Admin Eligibility Config (Admin)	BN-016	FR-003, FR-011	P2
US-22	Audit Trail	BN-008	FR-008	P0
US-23	Document Encryption + Access	BN-018	FR-006	P1
US-24	Bilingual UI (Arabic RTL)	BN-010	FR-010	P1
US-25	Program Lifecycle Management	BN-015	FR-012	P2
US-26	Duplicate Detection	BN-017	FR-003	P2

Total User Stories: 26 **P0 (Must Have — MVP)**: US-01, US-02, US-03, US-04, US-05, US-06, US-08, US-09, US-10, US-11, US-12, US-13, US-14, US-15, US-22 — **15 stories P1 (Should Have — MVP)**: US-16, US-17, US-18, US-19, US-20, US-23, US-24 — **7 stories P2 (Nice to Have — Phase 2)**: US-07, US-21, US-25, US-26 — **4 stories**

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